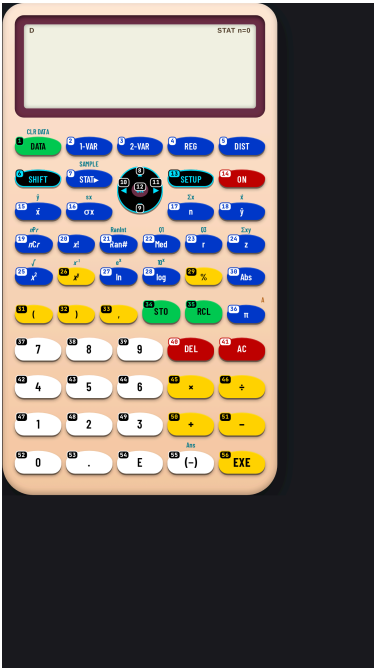
U-45 CALCULATORS

User Manual

Calculator #42 Statistics Data-Editor Scientific (orange)

Scientific – natural display



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#42 — Statistics Data-Editor Scientific (orange)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example [#1 DATA]. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

algebraic with a statistics data editor (enter data in DATA, then 1-VAR, 2-VAR, REG or DIST).

Chapters

1. Getting started
2. Features
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4. Numbered key map
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1 · Getting started

Open U-45 CALCULATORS, choose **#42** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#37, #38, #39, #42, #43, #44, #47, #48, #49, #52, #53, #54, #55
op	Operators & execute	#12, #26, #29, #31, #32, #33, #45, #46, #50, #51, #56
fn	Functions	#2, #3, #4, #5, #7, #15, #16, #17, #18, #19, #20, #21, #22, #23, #24, #25, #27, #28, #30, #36
mem	Memory, variables & registers	#1, #34, #35
clr	Clear, delete & power	#14, #40, #41

mod	**Shift, modes, menus & navigation**	**#6, #8, #9, #10, #11, #13**

2 · Features

45 features: 31 live on the keys, 9 app/UI features, 5 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE **Two-column X / Y data editor with row numbers, append, edit and delete**
2. LIVE **One-variable statistics: n, mean, Σx, Σx^2, sample and population standard deviation**
3. LIVE **Five-number summary: min, Q1, median, Q3, max, plus range and IQR**
4. LIVE **Mode (all most-frequent values)**
5. LIVE **Two-variable statistics: means, sums, Σxy, sx, sy, covariance and r**
6. LIVE **Linear regression $y = a + bx$ with r and r^2**
7. LIVE **Quadratic regression $y = a + bx + cx^2$**
8. LIVE **Exponential regression $y = a \cdot e^{(bx)}$**
9. LIVE **Power regression $y = a \cdot x^b$**
10. LIVE **Logarithmic regression $y = a + b \cdot \ln x$**
11. LIVE **Prediction $\hat{y}(x)$ from the last regression**
12. LIVE **Inverse prediction $\hat{x}(y)$**
13. LIVE **z-score of a value**
14. LIVE **Normal probability density**
15. LIVE **Normal cumulative probability between two bounds**
16. LIVE **Inverse normal**
17. LIVE **Binomial PD and CD**
18. LIVE **Poisson PD and CD**
19. LIVE **Geometric PD and CD**

20. LIVE Student-t cumulative probability and inverse t
21. LIVE χ^2 cumulative probability
22. LIVE Exponential and uniform cumulative probability
23. LIVE Statistics variables usable in formulas ($\bar{x}$, sx, σx, n, Σx, Σx^2, $\bar{y}$, Σxy, Med, Q1, Q3, r)
24. LIVE nCr and nPr combinatorics, n!
25. LIVE Random numbers: Ran#, RanInt, RanNorm(μ,σ), RanBin(n,p)
26. LIVE Sample data set for practice
27. LIVE Clear-data key
28. LIVE Distribution results go to Ans and the history
29. LIVE Scientific functions, powers, logs and memory
30. LIVE Two-line history with recall
31. LIVE SETUP formats Fix / Sci / Eng
32. APP/UI Every key numbered 1…N in reading order — matches the key map in the user manual
33. APP/UI High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a “Classic” toggle
34. APP/UI Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)
35. APP/UI Show / hide key numbers with one button (remembered between sessions)
36. APP/UI Copy the current display text to the clipboard
37. APP/UI Physical-keyboard shortcuts: digits, + − * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i
38. APP/UI Fit / 1:1 zoom that never pushes keys off a phone screen

39. APP/UI **Reference photo viewer and tool-tip name on every key**
40. APP/UI **Reset = fresh power-on state; works fully offline (PWA cache / Android APK)**
41. PLANNED **Exam mode (locks memory & shows indicator)**
42. PLANNED **Answer history export to text**
43. PLANNED **Speech read-out of results**
44. PLANNED **Colour-blind safe key theme**
45. PLANNED **Custom shortcut keys**

2b · Feature guide

Calculator #42 is a algebraic with a statistics data editor (enter data in DATA, then 1-VAR, 2-VAR, REG or DIST). Each card below is one of its 31 working features, with the keys that reach it and, where one exists, a verified example.

F1 Two-column X / Y data editor with row numbers, append, edit and delete

Where: #1 DATA Open the X/Y data editor; #7 STAT▶ Statistics variables and functions menu; #17 n Number of data

See it in action: Mean of entered data — *x̄ of 2, 4, 9 (DATA editor)* = 5; Sum of entered data — *Σx of 3, 5, 7* = 15

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F2 One-variable statistics: n, mean, Σx, Σx², sample and population standard deviation

Where: #2 1-VAR One-variable statistics of X; #3 2-VAR Two-variable statistics of X,Y; #7 STAT▶ Statistics variables and functions menu

See it in action: Mean of entered data — *x̄ of 2, 4, 9 (DATA editor)* = 5

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F3 Five-number summary: min, Q1, median, Q3, max, plus range and IQR

Where: #21 Ran# Random number 0–1; #13 SETUP SETUP — angle unit, number format, display options; #17 n Number of data

See it in action: Median — *Median of 8, 1, 5* = 5

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F4 Mode (all most-frequent values)

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F5 Two-variable statistics: means, sums, Σxy, sx, sy, covariance and r

Where: #2 1-VAR One-variable statistics of X; #3 2-VAR Two-variable statistics of X,Y; #7 STAT▶ Statistics variables and functions menu

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F6 Linear regression y = a + bx with r and r^2

Where: #4 REG Regression models; #18 ŷ Predict y from the regression

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F7 Quadratic regression y = a + bx + cx²

Where: #4 REG Regression models; #18 ŷ Predict y from the regression

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F8 Exponential regression y = a·e^(bx)

Where: #4 REG Regression models; #18 ŷ Predict y from the regression; #27 ln Natural logarithm ln

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F9 Power regression y = a·x^b

Where: #4 REG Regression models; #14 ON ON — power on; #18 ŷ Predict y from the regression

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F10 Logarithmic regression y = a + b·ln x

Where: #4 REG Regression models; #18 ŷ Predict y from the regression

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F11 Prediction ŷ(x) from the last regression

Where: #4 REG Regression models; #18 ŷ Predict y from the regression; #55 (−) Negative sign (−) — makes the next number negative. Different from the subtraction key

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F12 Inverse prediction x̂(y)

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F13 z-score of a value

Where: #24 z z-score

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F14 Normal probability density

Where: #5 DIST Probability distributions

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F15 Normal cumulative probability between two bounds

Where: #5 DIST Probability distributions; #33 , Comma / argument separator (between function arguments, list items or coordinates)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F16 Inverse normal

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F17 Binomial PD and CD

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F18 Poisson PD and CD

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F19 Geometric PD and CD

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F20 Student-t cumulative probability and inverse t

Where: #5 DIST Probability distributions

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F21 χ^2 cumulative probability

Where: #5 DIST Probability distributions

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F22 Exponential and uniform cumulative probability

Where: #5 DIST Probability distributions; #27 ln Natural logarithm ln

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F23 Statistics variables usable in formulas (x̄, sx, σx, n, Σx, Σx², ȳ, Σxy, Med, Q1, Q3, r)

Where: #7 STAT▶ Statistics variables and functions menu; #2 1-VAR One-variable statistics of X; #3 2-VAR Two-variable statistics of X,Y

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F24 nCr and nPr combinatorics, n!

Where: #19 nCr Combinations nCr

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F25 Random numbers: Ran#, RanInt, RanNorm(μ,σ), RanBin(n,p)

Where: #21 Ran# Random number 0–1

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F26 Sample data set for practice

Where: #7 STAT▶ Statistics variables and functions menu; #1 DATA Open the X/Y data editor; #16 σx Population s.d.

See it in action: Mean of entered data — *x̄ of 2, 4, 9 (DATA editor)* = 5; Sum of entered data — *Σx of 3, 5, 7* = 15

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F27 Clear-data key

Where: #1 DATA Open the X/Y data editor; #7 STAT▶ Statistics variables and functions menu; #17 n Number of data

See it in action: Mean of entered data — *x̄ of 2, 4, 9 (DATA editor)* = 5; Sum of entered data — *Σx of 3, 5, 7* = 15

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F28 Distribution results go to Ans and the history

Where: #8 ▲ Cursor ▲ — up / previous history line / previous item; #9 ▼ Cursor ▼ — down / next history line / next item; #55 (−) Negative sign (−) — makes the next number negative. Different from the subtraction key

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F29 Scientific functions, powers, logs and memory

Where: #7 STAT▶ Statistics variables and functions menu; #34 STO STO — store the value into a memory / variable (press the target next); #35 RCL RCL — recall a memory / variable

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F30 Two-line history with recall

Where: #8 ▲ Cursor ▲ — up / previous history line / previous item; #9 ▼ Cursor ▼ — down / next history line / next item; #10 ◀ Cursor ◀ — move left in the line

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F31 SETUP formats Fix / Sci / Eng

Where: #13 SETUP **SETUP — angle unit, number format, display options**

Try it: Press Reset, use the key(s) above, and write the display here:

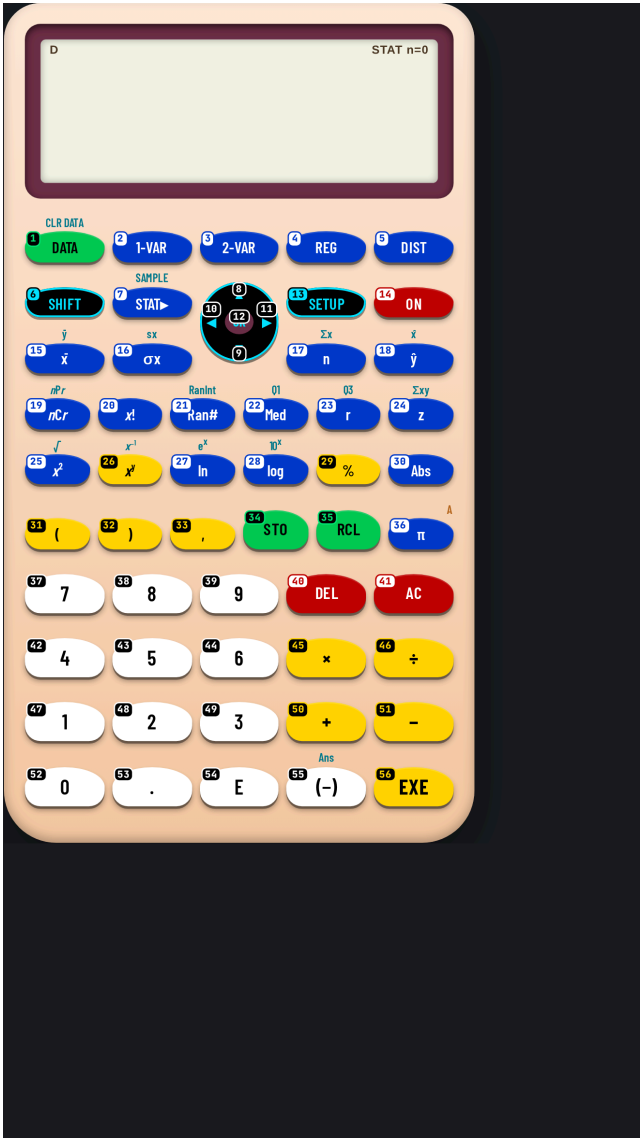
_____

Your notes:

_____

_____

3 · Numbered key map



Calculator #42 with key numbers shown (Hi-Contrast colours).

Key table

“Shift” = 2nd / f / SHIFT layer; “Alpha” = ALPHA layer; “3rd” = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	DATA	CLR DATA	·	·	mem
2	1-VAR	·	·	·	fn
3	2-VAR	·	·	·	fn
4	REG	·	·	·	fn
5	DIST	·	·	·	fn
6	SHIFT	·	·	·	mod
7	STAT▸	SAMPLE	·	·	fn
8	▲	·	·	·	mod
9	▼	·	·	·	mod
10	◀	·	·	·	mod
11	▶	·	·	·	mod
12	OK	·	·	·	op
13	SETUP	·	·	·	mod
14	ON	·	·	·	clr
15	x̄	ȳ	·	·	fn
16	σx	sx	·	·	fn
17	n	Σx	·	·	fn
18	ŷ	x̂	·	·	fn
19	nCr	nPr	·	·	fn
20	x!	·	·	·	fn
21	Ran#	RanInt	·	·	fn
22	Med	Q1	·	·	fn
23	r	Q3	·	·	fn

#	Face	Shift	Alpha	3rd	Group
24	z	Σxy	·	·	fn
25	x^2	√	·	·	fn
26	xy	x^{-1}	·	·	op
27	ln	ex	·	·	fn
28	log	10x	·	·	fn
29	%	·	·	·	op
30	Abs	·	·	·	fn
31	(	·	·	·	op
32	)	·	·	·	op
33	,	·	·	·	op
34	STO	·	·	·	mem
35	RCL	·	·	·	mem
36	π	·	A	·	fn
37	7	·	·	·	num
38	8	·	·	·	num
39	9	·	·	·	num
40	DEL	·	·	·	clr
41	AC	·	·	·	clr
42	4	·	·	·	num
43	5	·	·	·	num
44	6	·	·	·	num
45	×	·	·	·	op
46	÷	·	·	·	op
47	1	·	·	·	num
48	2	·	·	·	num
49	3	·	·	·	num
50	+	·	·	·	op
51	−	·	·	·	op

#	Face	Shift	Alpha	3rd	Group
52	0	·	·	·	num
53	.	·	·	·	num
54	E	·	·	·	num
55	(–)	Ans	·	·	num
56	EXE	·	·	·	op

4 · Key-by-key reference

Every key on calculator #42, in number order, with every function it carries in this app.

#1 DATA `Memory, variables & registers`

Primary: Open the X/Y data editor

shifted (2nd / f / SHIFT layer) (press #6 first): CLR DATA — Clear all data

#2 1-VAR `Functions`

Primary: One-variable statistics of X

#3 2-VAR `Functions`

Primary: Two-variable statistics of X,Y

#4 REG `Functions`

Primary: Regression models

#5 DIST `Functions`

Primary: Probability distributions

#6 SHIFT `Shift, modes, menus & navigation`

Primary: SHIFT / 2nd — the next key uses its upper (shift-coloured) function

#7 STAT▶ `Functions`

Primary: Statistics variables and functions menu

shifted (2nd / f / SHIFT layer) (press #6 first): SAMPLE — Load sample data

#8 ▲ `Shift, modes, menus & navigation`

Primary: Cursor ▲ — up / previous history line / previous item

#9 ▼ `Shift, modes, menus & navigation`

Primary: Cursor ▼ — down / next history line / next item

#10 ◀ `Shift, modes, menus & navigation`

Primary: Cursor ◀ — move left in the line

#11 ▶ `Shift, modes, menus & navigation`

Primary: Cursor ▶ — move right in the line

#12 OK `Operators & execute`

Primary: OK — confirm the highlighted item

#13 SETUP `Shift, modes, menus & navigation`

Primary: SETUP — angle unit, number format, display options

#14 ON Clear, delete & power

Primary: ON — power on

#15 $\bar{x}$ Functions

Primary: Mean of X

shifted (2nd / f / SHIFT layer) (press #6 first): $\bar{y}$ — Mean of Y

#16 σx Functions

Primary: Population s.d.

shifted (2nd / f / SHIFT layer) (press #6 first): sx — Sample s.d.

#17 n Functions

Primary: Number of data

shifted (2nd / f / SHIFT layer) (press #6 first): Σx — Sum of X

#18 $\hat{y}$ Functions

Primary: Predict y from the regression

shifted (2nd / f / SHIFT layer) (press #6 first): $\hat{x}$ — Predict x

#19 nCr Functions

Primary: Combinations nCr

shifted (2nd / f / SHIFT layer) (press #6 first): nPr — Permutations nPr

#20 x! Functions

Primary: Factorial n! (also on non-integers via the gamma function where supported)

Keyboard: !

#21 Ran# Functions

Primary: Random number 0–1

shifted (2nd / f / SHIFT layer) (press #6 first): RanInt — Random integer in a range

#22 Med Functions

Primary: Median

shifted (2nd / f / SHIFT layer) (press #6 first): Q1 — First quartile

#23 r Functions

Primary: Correlation coefficient

shifted (2nd / f / SHIFT layer) (press #6 first): Q3 — Third quartile

#24 z Functions

Primary: z-score

shifted (2nd / f / SHIFT layer) (press #6 first): Σxy — Sum of products

#25 x^2 Functions

Primary: Square x^2

shifted (2nd / f / SHIFT layer) (press #6 first): $\sqrt{}$ — Square root $\sqrt{}$

Keyboard: q

#26 xy Operators & execute

Primary: Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)

shifted (2nd / f / SHIFT layer) (press #6 first): x^{-1} — Reciprocal x^{-1} (1/x)

Keyboard: ^

#27 ln Functions

Primary: Natural logarithm ln

shifted (2nd / f / SHIFT layer) (press #6 first): ex — e^x — exponential

Keyboard: n

#28 log Functions

Primary: Common logarithm $\log_{10}$

shifted (2nd / f / SHIFT layer) (press #6 first): 10x — 10^x — antilog

Keyboard: l

#29 % Operators & execute

Primary: Percent %

Keyboard: %

#30 Abs Functions

Primary: Absolute value |x|

#31 (Operators & execute

Primary: Opening bracket (

Keyboard: (

#32) Operators & execute

Primary: Closing bracket). Closing brackets at the end of a line may be omitted

Keyboard:)

#33 , Operators & execute

Primary: Comma / argument separator (between function arguments, list items or coordinates)

Keyboard: ,

#34 STO Memory, variables & registers

Primary: STO — store the value into a memory / variable (press the target next)

#35 RCL Memory, variables & registers

Primary: RCL — recall a memory / variable

#36 π Functions

Primary: π (3.14159265…) as an exact constant

alpha layer: A — Variable A (store with STO/ →, recall by name)

Keyboard: p

#37 7 Digits & number entry

Primary: Digit 7

Keyboard: 7

#38 8 Digits & number entry

Primary: Digit 8

Keyboard: 8

#39 9 Digits & number entry

Primary: Digit 9

Keyboard: 9

#40 DEL Clear, delete & power

Primary: DEL — delete the character at the cursor

Keyboard: Backspace Delete

#41 AC `Clear, delete & power`

Primary: AC — clear all of the current calculation

Keyboard: Escape

#42 4 `Digits & number entry`

Primary: Digit 4

Keyboard: 4

#43 5 `Digits & number entry`

Primary: Digit 5

Keyboard: 5

#44 6 `Digits & number entry`

Primary: Digit 6

Keyboard: 6

#45 × `Operators & execute`

Primary: Multiplication ×

Keyboard: * x

#46 ÷ `Operators & execute`

Primary: Division ÷

Keyboard: /

#47 1 Digits & number entry

Primary: Digit 1

Keyboard: 1

#48 2 Digits & number entry

Primary: Digit 2

Keyboard: 2

#49 3 Digits & number entry

Primary: Digit 3

Keyboard: 3

#50 + Operators & execute

Primary: Addition +

Keyboard: +

#51 − Operators & execute

Primary: Subtraction −

Keyboard: -

#52 0 Digits & number entry

Primary: Digit 0

Keyboard: 0

#53 . Digits & number entry

Primary: Decimal point. Pressing it first gives "0."; a second point in the same number is ignored

Keyboard: .

#54 E Digits & number entry

Primary: ×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23

Keyboard: e

#55 (−) Digits & number entry

Primary: Negative sign (−) — makes the next number negative. Different from the subtraction key

shifted (2nd / f / SHIFT layer) (press #6 first): Ans — Ans — recall the last answer into the current expression

#56 EXE Operators & execute

Primary: Execute / = — evaluate the line and show the result (EXE / ENTER / =)

Keyboard: Enter =

5 · Worked examples

6 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Combinations

Calculate **10C3**. Press **Reset** first.

#47 1 → #52 0 → #19 *n*C*r* → #49 3 → #56 EXE

Display shows `120` — expected 120. ✔ verified on the app.

Example 5.2 — Permutations

Calculate **6P2**. Press **Reset** first.

#44 6 → #6 SHIFT → #19 *n*P*r* → #48 2 → #56 EXE

Display shows `30` — expected 30. ✔ verified on the app.

Example 5.3 — Mean of entered data

Calculate **x̄ of 2, 4, 9 (DATA editor)**. Press **Reset** first.

#1 DATA → #48 2 → #56 EXE → #42 4 → #56 EXE → #39 9 → #56 EXE → #41 AC → #15 x̄ → #56 EXE

Display shows `5` — expected 5. ✔ verified on the app.

Example 5.4 — Sum of entered data

Calculate **Σx of 3, 5, 7**. Press **Reset** first.

#1 DATA → #49 3 → #56 EXE → #43 5 → #56 EXE → #37 7 → #56 EXE → #41 AC → #6 SHIFT → #17 Σx → #56 EXE

Display shows `15` — expected 15. ✔ verified on the app.

Example 5.5 — Count of data

Calculate **n for 1, 1, 2, 3**. Press **Reset** first.

#1 DATA → #47 1 → #56 EXE → #47 1 → #56 EXE → #48 2 → #56 EXE → #49 3 → #56 EXE → #41 AC → #17 n → #56 EXE

Display shows `4` — expected 4. ✔ verified on the app.

Example 5.6 — Median

Calculate **Median of 8, 1, 5**. Press **Reset** first.

#1 DATA → #38 8 → #56 EXE → #47 1 → #56 EXE → #43 5 → #56 EXE → #41 AC → #22 Med → #56 EXE

Display shows `5` — expected 5. ✔ verified on the app.

6 · Tricks & hacks

Hack 1: Edit instead of retyping

After a result, press ◀ or ▶ to bring the last expression back with the cursor in it; change one number and press = again.

Hack 2: Ans chains

Start a line with an operator (e.g. ×) and the calculator inserts Ans automatically — great for compound growth: 1.05 ×, then = = = …

Hack 3: Exact vs decimal

Natural-display models keep fractions, √ and π exact. Press S⇔D to flip to decimal; press it again to return.

Hack 4: Close brackets for free

Trailing closing brackets can be left off; = closes them for you.

Hack 5: Implicit multiplication

Type 2π or 3(4+1) without ×. Implicit multiplication binds tighter than ÷ on textbook models — use brackets if in doubt.

Hack 6: Store constants

Store a value you reuse (e.g. g = 9.81) with STO and a letter; then type the letter in formulas.

Hack 7: Power chains

x^y chains right to left: 2^3^2 = 2^9 = 512, not 64. Use brackets: (2^3)^2 = 64.

Hack 8: Angles

Check the D / R / G indicator before trig. sin 30 = 0.5 only in degrees.

Hack 9: Memory sums

M+ / M− keep a running total separate from the calculation line; RCL M shows it.

Hack 10: Replay history

Use ▲ ▼ to scroll earlier lines; press = on an old line to re-run it.

Hack 11: Hidden functions on this model

17 extra functions sit on shifted layers. Press the layer key once, then the key: for example #6 then #1 → CLR DATA; #6 then #7 → SAMPLE; #6 then #15 → ȳ; #6 then #16 → sx.

Hack 12: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left → right, top → bottom from #1 to #56.

Hack 13: Type faster on a computer

29 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 14: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 15: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 16: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
!	#20	x!	Factorial n! (also on non-integers via the gamma function where supported)
q	#25	x^2	Square x^2
^	#26	xy	Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)
n	#27	ln	Natural logarithm ln
l	#28	log	Common logarithm $\log_{10}$
%	#29	%	Percent %
(	#31	(	Opening bracket (
)	#32	)	Closing bracket). Closing brackets at the end of a line may be omitted
,	#33	,	Comma / argument separator (between function arguments, list items or coordinates)
p	#36	π	π (3.14159265…) as an exact constant
7	#37	7	Digit 7
8	#38	8	Digit 8
9	#39	9	Digit 9
Backspace Delete	#40	DEL	DEL — delete the character at the cursor
Escape	#41	AC	AC — clear all of the current calculation
4	#42	4	Digit 4
5	#43	5	Digit 5
6	#44	6	Digit 6
* x	#45	×	Multiplication ×

/	#46	÷	Division ÷
1	#47	1	Digit 1
2	#48	2	Digit 2
3	#49	3	Digit 3
+	#50	+	Addition +
-	#51	–	Subtraction –
0	#52	0	Digit 0
.	#53	.	Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored
e	#54	E	×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23
Enter =	#56	EXE	Execute / = — evaluate the line and show the result (EXE / ENTER / =)

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: Using – instead of (–)

The negative key (–) and the subtraction key are different; "– 5" at the start uses Ans – 5.

Mistake 2: Wrong angle unit

If sin 30 gives –0.988, the calculator is in radians. Switch to degrees in SETUP.

Mistake 3: -3^2 vs $(-3)^2$

$-3^2 = -9$ because the square is applied first. Type $(-3)^2$ to get 9.

Mistake 4: Division precedence

$1 \div 2\pi$ may be read as $1 \div (2\pi)$ on textbook models. Type $1 \div (2 \times \pi)$ explicitly to be sure.

Mistake 5: Forgetting SHIFT is one-shot

SHIFT applies only to the next key. If the S indicator is still lit, press SHIFT again to cancel.

Mistake 6: Clearing memory by accident

AC clears the line but not memory; CLR (SHIFT 9) can clear memory and variables.

Mistake 7: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 8: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

Mistake 9: Minus vs negative

The (−) negative key and the − subtraction key are different. Use (−) to start a negative number, − between two numbers.

9 · Practice drills

Set A — Calculation drills (6)

Work each one on calculator #42, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. 10C3 = _____
2. 6P2 = _____
3. $\bar{x}$ of 2, 4, 9 (DATA editor) = _____
4. Σx of 3, 5, 7 = _____
5. n for 1, 1, 2, 3 = _____
6. Median of 8, 1, 5 = _____

Set B — Key-finding drills (63)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. Which key is “DATA” (Open the X/Y data editor)? _____
2. Which key is “1-VAR” (One-variable statistics of X)? _____
3. Which key is “2-VAR” (Two-variable statistics of X,Y)? _____
4. Which key is “REG” (Regression models)? _____
5. Which key is “DIST” (Probability distributions)? _____
6. Which key is “SHIFT” (SHIFT / 2nd)? _____
7. Which key is “STAT▸” (Statistics variables and functions menu)? _____
8. Which key is “ ▲ ” (Cursor ▲)? _____

9. Which key is " ▼ " (Cursor ▼)? _____

10. Which key is "◀" (Cursor ◀)? _____

11. Which key is "▶" (Cursor ▶)? _____

12. Which key is "OK" (OK)? _____

13. Which key is "SETUP" (SETUP)? _____

14. Which key is "ON" (ON)? _____

15. Which key is "$\bar{x}$" (Mean of X)? _____

16. Which key is "σx" (Population s.d.)? _____

17. Which key is "n" (Number of data)? _____

18. Which key is "$\hat{y}$" (Predict y from the regression)? _____

19. Which key is "nCr" (Combinations nCr)? _____

20. Which key is "x!" (Factorial n!)? _____

21. Which key is "Ran#" (Random number 0–1)? _____

22. Which key is "Med" (Median)? _____

23. Which key is "r" (Correlation coefficient)? _____

24. Which key is "z" (z-score)? _____

25. Which key is "x^2" (Square x^2)? _____

26. Which key is "xy" (Power x^y / ^)? _____

27. Which key is "ln" (Natural logarithm ln)? _____

28. Which key is "log" (Common logarithm $\log_{10}$)? _____

29. Which key is “%” (Percent %)? _____

30. Which key is “Abs” (Absolute value |x|)? _____

31. Which key is “(” (Opening bracket)? _____

32. Which key is “)” (Closing bracket). Closing brackets at the end of a line may be omitted)? _____

33. Which key is “,” (Comma / argument separator)? _____

34. Which key is “STO” (STO)? _____

35. Which key is “RCL” (RCL)? _____

36. Which key is “π” (π)? _____

37. Which key is “DEL” (DEL)? _____

38. Which key is “AC” (AC)? _____

39. Which key is “×” (Multiplication ×)? _____

40. Which key is “÷” (Division ÷)? _____

41. Which key is “+” (Addition +)? _____

42. Which key is “−” (Subtraction −)? _____

43. Which key is “.” (Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored)? _____

44. Which key is “E” (×10ˣ / EXP / EE)? _____

45. Which key is “(−)” (Negative sign)? _____

46. Which key is “EXE” (Execute / =)? _____

47. How do you reach “CLR DATA”? _____

48. How do you reach “SAMPLE”? _____

49. How do you reach “$\bar{y}$”? _____

50. How do you reach “sx”? _____

51. How do you reach “Σx”? _____

52. How do you reach “$\hat{x}$”? _____

53. How do you reach “nPr”? _____

54. How do you reach “RanInt”? _____

55. How do you reach “Q1”? _____

56. How do you reach “Q3”? _____

57. How do you reach “Σxy”? _____

58. How do you reach “$\sqrt{\ }$”? _____

59. How do you reach “x^{-1}”? _____

60. How do you reach “ex”? _____

61. How do you reach “10x”? _____

62. How do you reach “A”? _____

63. How do you reach “Ans”? _____

Appendix A — Drill answers

Set A

1. 120 (display `120`)

`#47` 1 → `#52` 0 → `#19` *n*C*r* → `#49` 3 → `#56` EXE

2. 30 (display `30`)

`#44` 6 → `#6` SHIFT → `#19` *n*P*r* → `#48` 2 → `#56` EXE

3. 5 (display `5`)

`#1` DATA → `#48` 2 → `#56` EXE → `#42` 4 → `#56` EXE → `#39` 9 → `#56` EXE → `#41` AC → `#15` $\bar{x}$ → `#56` EXE

4. 15 (display `15`)

`#1` DATA → `#49` 3 → `#56` EXE → `#43` 5 → `#56` EXE → `#37` 7 → `#56` EXE → `#41` AC → `#6` SHIFT → `#17` Σx → `#56` EXE

5. 4 (display `4`)

`#1` DATA → `#47` 1 → `#56` EXE → `#47` 1 → `#56` EXE → `#48` 2 → `#56` EXE → `#49` 3 → `#56` EXE → `#41` AC → `#17` n → `#56` EXE

6. 5 (display `5`)

`#1` DATA → `#38` 8 → `#56` EXE → `#47` 1 → `#56` EXE → `#43` 5 → `#56` EXE → `#41` AC → `#22` Med → `#56` EXE

Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #14
15. #15
16. #16
17. #17
18. #18
19. #19
20. #20
21. #21
22. #22
23. #23
24. #24
25. #25
26. #26
27. #27
28. #28
29. #29
30. #30
31. #31
32. #32
33. #33
34. #34
35. #35
36. #36
37. #40
38. #41
39. #45
40. #46
41. #50
42. #51
43. #53
44. #54
45. #55
46. #56
47. #6 SHIFT then #1 DATA
48. #6 SHIFT then #7 STAT▸
49. #6 SHIFT then #15 $\bar{x}$
50. #6 SHIFT then #16 σx
51. #6 SHIFT then #17 n
52. #6 SHIFT then #18 ŷ
53. #6 SHIFT then #19 nCr
54. #6 SHIFT then #21 Ran#
55. #6 SHIFT then #22 Med
56. #6 SHIFT then #23 r
57. #6 SHIFT then #24 z
58. #6 SHIFT then #25 x^2
59. #6 SHIFT then #26 xy
60. #6 SHIFT then #27 ln
61. #6 SHIFT then #28 log
62. #36 (alpha layer)
63. #6 SHIFT then #55 (-)

Appendix B — Function finder (A–Z)

Function	Label	Keys
×10^x / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10^x 23	E	#54
10^x — antilog	10x	#6 → #28
Absolute value |x|	Abs	#30
AC — clear all of the current calculation	AC	#41
Addition +	+	#50
Ans — recall the last answer into the current expression	Ans	#6 → #55
Clear all data	CLR DATA	#6 → #1
Closing bracket). Closing brackets at the end of a line may be omitted	)	#32
Combinations nCr	nCr	#19
Comma / argument separator (between function arguments, list items or coordinates)	,	#33
Common logarithm $\log_{10}$	log	#28
Correlation coefficient	r	#23
Cursor ▲ — up / previous history line / previous item	▲	#8
Cursor ▶ — move right in the line	▶	#11
Cursor ▼ — down / next history line / next item	▼	#9
Cursor ◀ — move left in the line	◀	#10
Decimal point. Pressing it first gives "0."; a second point in the same number is ignored	.	#53
DEL — delete the character at the cursor	DEL	#40
Digit 0	0	#52
Digit 1	1	#47
Digit 2	2	#48

Digit 3	3	#49
Digit 4	4	#42
Digit 5	5	#43
Digit 6	6	#44
Digit 7	7	#37
Digit 8	8	#38
Digit 9	9	#39
Division ÷	÷	#46
e^x — exponential	ex	#6 → #27
Execute / = — evaluate the line and show the result (EXE / ENTER / =)	EXE	#56
Factorial n! (also on non-integers via the gamma function where supported)	x!	#20
First quartile	Q1	#6 → #22
Load sample data	SAMPLE	#6 → #7
Mean of X	$\bar{x}$	#15
Mean of Y	$\bar{y}$	#6 → #15
Median	Med	#22
Multiplication ×	×	#45
Natural logarithm ln	ln	#27
Negative sign (−) — makes the next number negative. Different from the subtraction key	(−)	#55
Number of data	n	#17
OK — confirm the highlighted item	OK	#12
ON — power on	ON	#14
One-variable statistics of X	1-VAR	#2
Open the X/Y data editor	DATA	#1
Opening bracket (	(	#31

Percent %	%	#29
Permutations nPr	nPr	#6 → #19
Population s.d.	σx	#16
Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)	xy	#26
Predict x	x̂	#6 → #18
Predict y from the regression	ŷ	#18
Probability distributions	DIST	#5
Random integer in a range	RanInt	#6 → #21
Random number 0–1	Ran#	#21
RCL — recall a memory / variable	RCL	#35
Reciprocal x^{-1} (1/x)	x^{-1}	#6 → #26
Regression models	REG	#4
Sample s.d.	sx	#6 → #16
SETUP — angle unit, number format, display options	SETUP	#13
SHIFT / 2nd — the next key uses its upper (shift-coloured) function	SHIFT	#6
Square root √	√	#6 → #25
Square x^2	x^2	#25
Statistics variables and functions menu	STAT▸	#7
STO — store the value into a memory / variable (press the target next)	STO	#34
Subtraction –	–	#51
Sum of products	Σxy	#6 → #24
Sum of X	Σx	#6 → #17

Third quartile	Q3	#6 → #23
Two-variable statistics of X,Y	2-VAR	#3
Variable A (store with STO/ →, recall by name)	A	#36
z-score	z	#24
π (3.14159265…) as an exact constant	π	#36

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

Statistics & probability calculator with an on-screen data editor. Sample standard deviation sx uses n−1; σx uses n. Quartiles use the median-of-halves method.

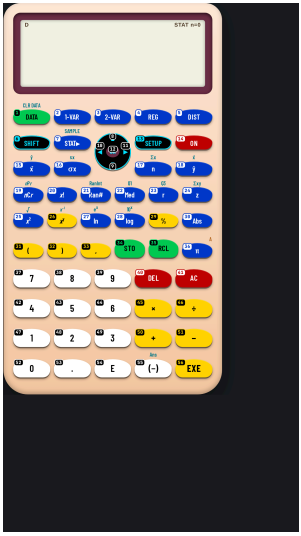
Distribution functions are original numerical code (series, continued fractions and Simpson integration) accurate to about 9 significant digits.

Planned (not yet working)

- Exam mode (locks memory & shows indicator)
- Answer history export to text
- Speech read-out of results
- Colour-blind safe key theme
- Custom shortcut keys

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 DATA ⇧ CLR DATA	#2 1-VAR	#3 2-VAR
#4 REG	#5 DIST	#6 SHIFT
#7 STAT▸ ⇧ SAMPLE	#8 ▲	#9 ▼
#10 ◀	#11 ▶	#12 OK
#13 SETUP	#14 ON	#15 $\bar{x}$ ⇧ $\bar{y}$
#16 σx ⇧ sx	#17 n ⇧ Σx	#18 $\hat{y}$ ⇧ $\hat{x}$
#19 nCr ⇧ nPr	#20 x!	#21 Ran# ⇧ RanInt
#22 Med ⇧ Q1	#23 r ⇧ Q3	#24 z ⇧ Σxy
#25 x^2 ⇧ √	#26 xy ⇧ x^{-1}	#27 ln ⇧ ex
#28 log ⇧ 10x	#29 %	#30 Abs
#31 (	#32)	#33 ,
#34 STO	#35 RCL	#36 π
#37 7	#38 8	#39 9
#40 DEL	#41 AC	#42 4
#43 5	#44 6	#45 ×
#46 ÷	#47 1	#48 2
#49 3	#50 +	#51 −

#52 0	**#53** .	**#54** E
#55 (−) ⇧ Ans	**#56** EXE	

U-45 CALCULATORS · Calculator #42 · cut out and keep